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The role of hydrologic information in reservoir operation – Learning from historical releases

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ABSTRACT

Closing the gap between theoretical reservoir operation and the real-world implementation remains a challenge in contemporary reservoir operations. Past research has focused on optimization algorithms and establishing optimal policies for reservoir operations. In this study, we attempt to understand operators' release decisions by investigating historical release data from 79 reservoirs in California and the Great Plains, using a data-mining approach. The 79 reservoirs are classified by hydrological regions, intra-annual seasons, average annual precipitation (climate), ratio of maximum reservoir capacity to average annual inflow (size ratio), hydrologic uncertainty associated with inflows, and reservoirs' main usage. We use information theory – specifically, mutual information – to measure the quality of inference between a set of classic indicators and observed releases at the monthly and weekly timescales. Several general trends are found to explain which sources of hydrologic information dictate reservoir release decisions under different conditions. Current inflow is the most important indicator during wet seasons, while previous releases are more relevant during dry seasons and in weekly data (as compared with monthly data). Inflow forecasting is the least important indicator in release decision making, but its importance increases linearly with hydrologic uncertainty and decreases logarithmically with reservoir size. No single hydrologic indicator is dominant across all reservoirs in either of the two regions.

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1. Introduction

Reservoir optimization is by far one of the most studied subjects in the water resources arena. Yet we still face the challenge of operating reservoirs in the real world by using the output of optimization models. Since the classic work of Young [39], the concept of establishing an optimal operation policy has become a central objective of enormous subsequent studies. Many advances in algorithms and computational efficiency (e.g., Dynamic Programming, Stochastic Dynamic Programming, Genetic Algorithms, etc.) have been accomplished and we are able to tackle multi-reservoir problems that were computationally prohibitive a few decades ago. Yet to date, optimal solutions find little appliance in the real world and a gap still exists between the endeavor of the water resources research community and reservoir operators [13,29,35,37].

Labadie [13] pointed out in the most recent review on optimal operations of multi-reservoir systems that many optimization models are unable to incorporate risk and uncertainty; but many reservoir managers are sensitive to hydrologic uncertainty and risk-aversion might be their main decision criterion. In this paper we further speculate that reservoir operators rely on some hydrologic information that is not fully considered in optimization models.

The importance of hydrologic information to the optimization of reservoirs was recognized early on as illustrated by Loucks et al. [14]. Past inflows, current inflows, forecasted inflows are some common hydrologic variables used in reservoir optimization models. Since these variables are posed as state variables in stochastic inflow modeling, they can be a cumbersome contribution to the computational burden of the reservoir optimization model; thus, generally only one hydrologic variable is used. Tejada-Guibert et al. [33] compared three different hydrologic variables in stochastic dynamic models of a multi-reservoir system in California and concluded that policies that employed more complete hydrologic information performed significantly better. They arrived at their findings by establishing several models and comparing improvements in model performance under predefined objective functions and differing sources of hydrologic information. Compared to their modeling perspective, here, we promote a data-mining approach to extract knowledge from historical release decisions and to learn which of such hydrologic variables were most relevant to reservoirs operations

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and under what conditions or criteria. Also to date, no rigorous effort has been undertaken to understand how release decisions were made by analyzing reservoir release records.

We utilize the concept of information theory to investigate the driving forces of release decisions in 79 reservoirs in California and the Great Plains. We test against several hydrologic variables to explore which sources of hydrologic information dictated decisions and under what circumstances. Although this paper may lead to the improvement of reservoir operation models, its central aim is to understand the human behavior of reservoir operators in the face of many factors with particular attention to the value of hydrologic information and uncertainty.

The rest of the paper is organized in the following order: we present the background of this study including a review of some relevant previous studies and an introduction of the concept of Shannon entropy, hydrologic uncertainty, and mutual information; describe the data used in the study; describe the methodology; present and discuss the results, and finally summarize the findings of the study.

2. Background

2.1. Previous relevant efforts

The idea of this work was initially inspired by the work of Howitt et al. [7], in which the authors raised the concept of calibrating a reservoir optimization problem based on historical reservoir release data to retrieve the risk-adverse contribution to the reservoir objective function. Although our work is not related to extracting the form of a risk-adverse function, it follows the premise of learning from past decisions to pose rather a different research question: Can we learn from historical records of reservoir releases, how reservoirs were operated and what factors dictated the decision process of reservoirs' operators? Interestingly, reservoir operation is probably one of the most studied subjects in the water research arena, yet we are not aware of any paper that has been directed to study observed releases except the works of Howitt et al. [7] and Barbagallo et al. [2]. Barbagallo et al. [2] extracted the operating rules of an irrigation reservoir in Sicily, Italy. They used Rough Set theory as a means to construct decision rules based on historical records of monthly released volumes during the irrigation season. They further claimed the suitability of the method as a knowledge discovery tool and suggested further incorporating the uncertainty associated with inflows.

In this paper, we build on the idea of Barbagallo et al. [2] but utilize a different methodological approach (information theory) to quantify hydrologic uncertainty and investigate operation rules from 79 selected reservoirs. The reservoirs will be further classified with respect to a set of reservoir parameters to extract general knowledge on how reservoirs were operated from a data-based perspective.

2.2. Shannon entropy and hydrologic uncertainty

Information theory and entropy have received considerable attention in hydrology and water resources. Entropy has been used in derivation of hydrologic probability distributions and parameter estimation [32], subsurface hydrogeologic simulation [3], hydrologic forecasting [22], classification of flow regimes [12], characterizing rainfall disorder [16], and designing monitoring networks [17,21,36]. To learn more about previous applications of entropy in hydrology and water resources, readers can be referred to Singh [30], Singh [31] and the references therein.

Shannon entropy (H) is defined as a measure of the level of uncertainty of a random variable. It also indicates how much information is needed on average to describe a random variable. Assuming X is a discrete random variable with alphabet $\mathbb{k}$ and probability mass function of $p(x) = \Pr\{X = x\}, x \in \mathbb{k}$, Shannon entropy can be defined as

$$H(X) = -\sum_{x\in\mathbb{k}} p(x)\cdot \log_n p(x) \tag{1}$$

where n is the logarithm base. In this study $n = 2$; thus, uncertainty is measured in units of bits.

Hydrologic uncertainty is quantified following the works of Amorocho and Espildora [1] and Chapman [6]. Hence, hydrologic uncertainty (HU) in a reservoir system can be defined as the entropy value of reservoir inflow (Q); e.g., HU = $H(Q)$. Similarly, the entropy of the observed release (R) data is a measure of decision uncertainty (DU), e.g., DU = $H(R)$. Higher values of entropy correspond to more uncertainty associated with the variable's measured state. In this paper, higher values of hydrologic uncertainty indicate higher uncertainty in reservoir inflow.

2.3. Mutual information

Due to the fact that reservoirs are stochastic, dynamic and potentially nonlinear systems, we chose to employ mutual information – a nonlinear technique – to extract relations between reservoir releases and several hydrologic variables. Mutual information is based on the classic work of Shannon [25] and was successfully employed as a nonlinear measure of inference among variables by many researchers (e.g. [10,11,18,19,23,26–28]).

Mutual information is defined as a measure of the amount of information that one random variable (the independent variable) contains or explains about another random variable (the dependent variable). It can also be explained as the amount of uncertainty reduction in the dependent variable due to the knowledge of the state of the independent variable. Suppose X and Y are random variables with a joint probability mass function $p(x,y)$ and marginal probability mass functions of $p(x)$ and $p(y)$, mutual information between X and Y can be defined as

$$I(X,Y) = \sum_{x\in\mathbb{k}}\sum_{y\in\mathbb{Z}} p(x,y)\cdot \log_2 \frac{p(x,y)}{p(x)\cdot p(y)} \tag{2}$$

Note, I ranges between 0 and ∞. $I(X,Y) = 0$ if and only if X and Y are two independent random variables (e.g., $p(x,y) = p(x)\cdot p(y)$, x and y represent realizations of X and Y). On the other hand, when the joint probability mass function $p(x,y)$ becomes greater than the product of the two marginal probability mass functions (e.g., $p(x)\cdot p(y)$), the value of I increases, indicating a strong dependence between the two variables. In order to compare reservoirs with different levels of uncertainty, we normalize the mutual information with respect to the marginal entropy of the dependent variable Y (Eq. (3)).

$$I_r(X,Y) = \frac{I(X,Y)}{H(Y)} \tag{3}$$

Thus, I_r is a value that ranges between 0 and 1. If $I_r = 0$ then there is no relationship between the two variables; but if $I_r = 1$ then the independent random variable x fully explains the variability in the dependant variable y. Unlike correlation coefficients, mutual information can capture nonlinear relevance between two variables, and is therefore more meaningful than linear correlations for potentially nonlinear datasets (e.g. [11,18]).

To determine whether the mutual information value between two random variables is statistically significant or not, we follow a standard shuffled-surrogate approach as implemented by Ruddell and Kumar [23]. This shuffled-surrogate method is used to establish statistical significance for non-parametric statistical methods applied to time series datasets, basing significance on a

surrogate statistic computed from a time-shuffled dataset with the same distribution as the original dataset.

Given time series datasets X and Y for which $I(X,Y)$ has already been computed, the values of X are shuffled randomly in time, maintaining the distribution of values in X but randomizing their time-lagged relationship to the values of Y. The mutual information $I_{\text{shuffled}}(X,Y)$ is computed for the shuffled X and original Y time series. This process is repeated many times, resulting in a Gaussian distribution for I_{shuffled} from which a mean and standard deviation of shuffled surrogates may be computed. Using this surrogate mean and standard deviation, the original I is tested against the hypothesis that it is indistinguishable from the surrogate I_{shuffled} at some chosen level of confidence using the traditional statistical hypothesis-testing framework. If I is found to be statistically distinguishable from I_{shuffled} at the chosen confidence level, the mutual information $I(X,Y)$ is said to be statistically significant. The shuffled-surrogate significance testing framework for entropy statistics has been applied by Kantz and Schurmann [8], Marschinski and Kantz [15], Sabesan et al. [24], Nichols [20], and others.

3. Reservoir data

Due to unavailability of historical release data for many reservoirs on one hand and excessive incomplete data on another hand, we were able to collect data for only 79 reservoirs in the US 22 of the reservoirs are located in the state of California with weekly records between 1963 and 2007 [5] and 57 reservoirs are in the Great Plains region (Colorado, Kansas, Montana, Nebraska, North Dakota, Oklahoma, South Dakota, Texas, & Wyoming) with monthly records between 1930 and 2007 [4]. Analyzing the 22 reservoirs in California at the monthly scale is excluded because the monthly time series are not long enough to reliably determine mutual information for the majority of the reservoirs. Fig. 1 provides a map of the locations of the selected reservoirs. Records of reservoir inflows (Q), reservoir releases (R), and reservoir storages (S) are collected for the reservoirs with at least 10 years of records. Also, average annual precipitation, maximum storage capacity, average annual inflow, main reservoir usage (e.g., flood control, irrigation, hydropower, or water supply) are compiled for each of the reservoirs.

In this study, we assume that reservoir capacities remain approximately constant over the duration of the records. This assumption is legitimate for two reasons. First, reservoir capacity changing annually is not available for any of the selected reservoirs. Second,. the reservoir size ratio parameter is employed to distinguish between large and small reservoirs relative to the amount of inflow they receive annually on average; a small discrepancy in the reservoir storage capacity would not turn a small size ratio to a large one or vice versa.

4. Methodology

4.1. Sources of hydrologic information

Reservoir operation models typically incorporate hydrologic information in the form of past inflows, current inflows, forecasted inflows, past storages, or past releases. In reality, however, human reservoir operators maybe rely on all of them or a combination of some of them under certain circumstances. Since weekly and monthly decisions may rely more on distant hydrologic information beyond the immediate past or future week or month, respectively, we determine the mutual information between reservoir release (R_t) and 17 hydrologic variables for the weekly data and 9 hydrologic variables for the monthly data as shown in Fig. 2. These variables are aggregated into five consistent variables for both the weekly and monthly data sets by selecting the maximum mutual information among the past and future four weeks (for weekly data) and past and future two months (for monthly data) (see Fig. 2), where I_r^Q, $I_r^{Q^-}$, $I_r^{Q^+}$, $I_r^{S^-}$, and $I_r^{R^-}$ are measures of the importance of current inflows, past inflows, future inflows, past storages and past releases, respectively, to current release. The aggregation of the hydrologic variables into the five consistent variables conform with the scope of the objective of this study – investigating

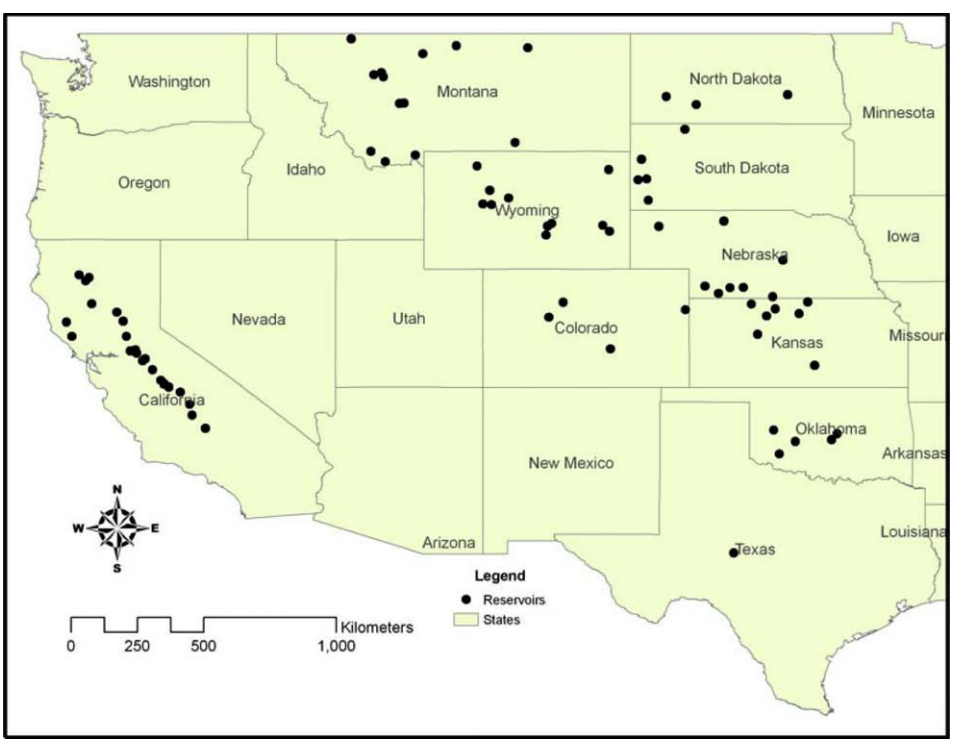

Fig. 1. Location of the 79 reservoirs.

Weekly Data	*Monthly Data*
1. $I_r^{R^-} = \text{Max}\left[I_r(R_{t-1}, R_t), I_r(R_{t-2}, R_t), I_r(R_{t-3}, R_t), I_r(R_{t-4}, R_t)\right]$	1. $I_r^{R^-} = \text{Max}\left[I_r(R_{t-1}, R_t), I_r(R_{t-2}, R_t)\right]$
2. $I_r^{S^-} = \text{Max}\left[I_r(S_{t-1}, R_t), I_r(S_{t-2}, R_t), I_r(S_{t-3}, R_t), I_r(S_{t-4}, R_t)\right]$	2. $I_r^{S^-} = \text{Max}\left[I_r(S_{t-1}, R_t), I_r(S_{t-2}, R_t)\right]$
3. $I_r^{Q^-} = \text{Max}\left[I_r(Q_{t-1}, R_t), I_r(Q_{t-2}, R_t), I_r(Q_{t-3}, R_t), I_r(Q_{t-4}, R_t)\right]$	3. $I_r^{Q^-} = \text{Max}\left[I_r(Q_{t-1}, R_t), I_r(Q_{t-2}, R_t)\right]$
4. $I_r^{Q^+} = \text{Max}\left[I_r(Q_{t+1}, R_t), I_r(Q_{t+2}, R_t), I_r(Q_{t+3}, R_t), I_r(Q_{t+4}, R_t)\right]$	4. $I_r^{Q^+} = \text{Max}\left[I_r(Q_{t+1}, R_t), I_r(Q_{t+2}, R_t)\right]$
5. $I_r^{Q} = \left[I_r(Q_t, R_t)\right]$	5. $I_r^{Q} = \left[I_r(Q_t, R_t)\right]$

Fig. 2. Aggregated sources of hydrologic information.

the importance of hydrologic information in the past, current, and future to release decisions – while easing the interpretation of results. For simplicity we refer to the five mutual information variables (I_r^Q, $I_r^{Q^-}$, $I_r^{Q^+}$, $I_r^{S^-}$, and $I_r^{R^-}$) through out the rest of the paper as Q, Q^-, Q^+, S^-, and R^-, respectively.

It should be noted that Q is a measure of a mix of immediate future and immediate past inflows. From a modeling point of view, Q is not truly known at time t and is considered forecasted information. However, from a decision making perspective, reservoir releases are usually not predetermined at the beginning of the period and left constant to the end of the decision horizon [38] – especially for longer temporal scales such as monthly operations. Rather decision making is a continuous process over the entire duration of the period and reservoir operators update the release function based on newly acquired hydrologic information [34]. Fig. 3 shows a schematic to explain the nature of hydrologic information during period t from the decision making perspective. It is supposed that t^* is the time at which a release decision is made between time t and $t + 1$; if $t^* = t$ then the content of the current period is solely immediate forecast information. If $t^* = t + 1$ then the content of the current period is solely immediate past information. If t^* is a value between t and $t + 1$, then inflows during the period (t to t^*) are viewed as past inflows while the inflows during the period (t^* to $t + 1$) are viewed as future inflow from the decision makers' standpoint. The complexity of Q differs with the temporal scale, i.e., the decision horizon. For example, for shorter intervals (e.g., weekly), Q includes more intermediate past and future information; for longer intervals (e.g., monthly), Q can include the information that is further away from the "current time point", such as the information in the past or future two weeks, as well as the immediate information.

4.2. Reservoir parameters for analysis

The 79 reservoirs are divided into two distinct groups on the basis of two reasons: they describe two different hydroclimatic regions (California verses Great Plains) and they have different temporal scales (weekly for reservoirs in California and monthly for reservoirs in the Great Plains). Furthermore, given the fact that flow is temporally variable and reservoirs are typically operated to serve different purposes during the different seasons, we split the data into seasons to further investigate variations in reservoir operations across seasons. Since snow melting could delay runoff substantially, high and low flow seasons don't necessarily coincide with wet and dry seasons. Fig. 4 shows the percentages of average reservoir inflow across seasons in both California and the Great Plains. Furthermore, in California, wet seasons are mainly during the winter and spring while summer and fall are typically dry; and in the Great Plains Region, the wet seasons are mainly during the spring and summer while fall and winter are typically dry. In both regions, spring and summer are the high flow seasons (Fig. 4).

The reservoirs are further classified on several additional parameters (see Fig. 5). Reservoirs are classified into five categories based on the most important source of hydrologic information (Q,

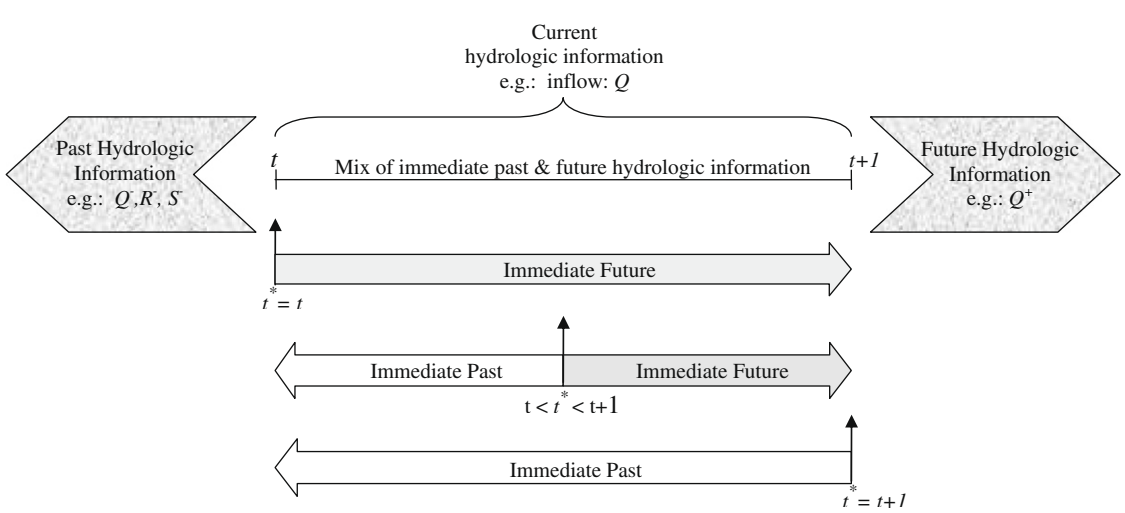

Fig. 3. Schematic illustrating the interpretation of current hydrologic information (specifically current inflow) from a decision perspective; t^* denotes the time a reservoir release decision is made within the temporal horizon of the decision period (e.g., weekly or monthly).

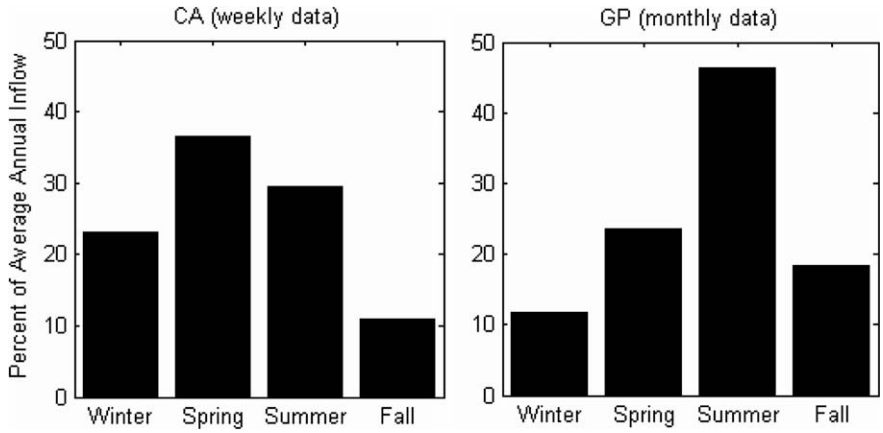

Fig. 4. Seasonal distribution of average annual inflows for the selected reservoirs in California (CA) and the Great Plains (GP).

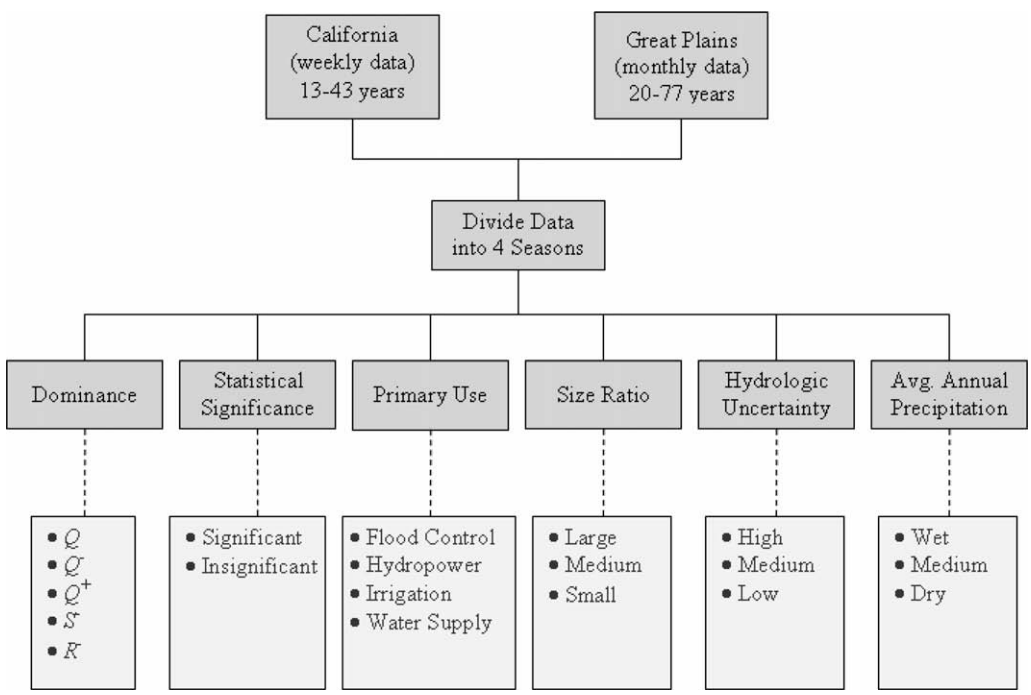

Fig. 5. A schematic of the classifications employed in the study; four seasons are defined as: winter (Nov–Jan), spring (Feb–Apr), summer (May–Jul), and fall (Aug–Oct); the statistical significance threshold value ($I_{\text{significance}}$) is established with a 99.5% significance level.

Q^-, Q^+, S^-, or R^-) to decision making. Reservoirs that exhibit statistically significant relations between reservoir release and a hydrologic variable are compared to those that do not. The significance threshold value ($I_{\text{significance}}$) is established with a very stringent 99.5% significance level. Reservoirs are also classified by their primary use, size ratio, and hydrologic uncertainty. Reservoir size ratio is defined as the ratio of the maximum reservoir capacity to average annual inflow of each reservoir; a large size ratio is indicative of a large reservoir size in a relative sense to its annual received inflow. Reservoirs with high hydrologic uncertainty indicate more highly random and unpredictable inflows. Lastly, the average annual precipitation associated with each reservoir is compiled to distinguish between reservoirs' operations in wet verses dry climatic regions. Such information can allow us to address if hydroclimatic conditions played any role in decision making of reservoir operators.

Prior to computing the mutual information values associated with each of the five hydrologic variables (Q, Q^-, Q^+, S^-, or R^-), the data records for each reservoir are divided into four seasons. The magnitudes of mutual information with respect to Q, Q^-, Q^+, S^-, and R^- are analyzed and compared across seasons and between the two regions. The results are further analyzed with respect to: single most important source of hydrologic information, statistical significance threshold, reservoirs' main usage, size ratio, hydrologic uncertainty, and average annual precipitation. Relations that emerge between any of the reservoir parameters and release decisions across the selected reservoirs are highlighted and discussed in the following section.

5. Results

On average, nonlinear statistics reveal strong ties between release decisions and hydrologic information – especially with current inflows (Q) and previous release (R^-).

That is to say reservoir operators have relied mainly on current inflow and previous release as to guide their operation decisions.

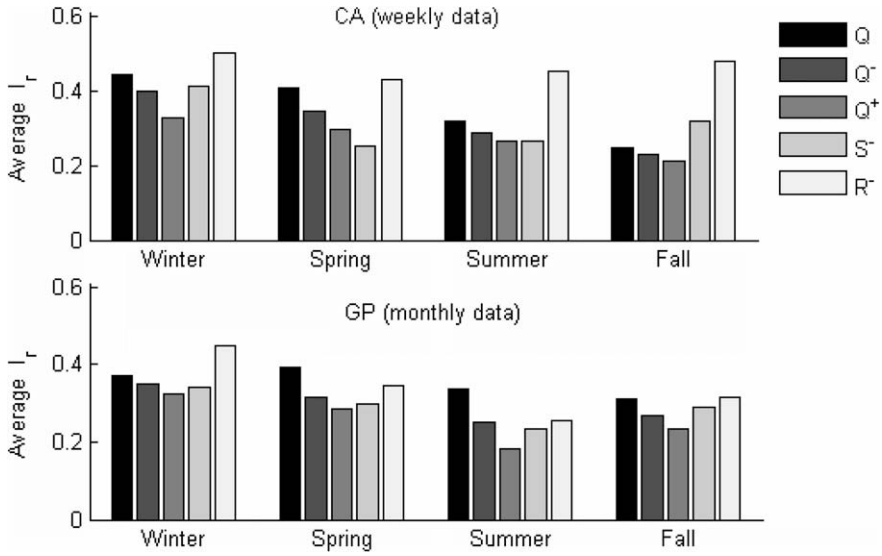

Fig. 6. Average relative mutual information associated with each of the five sources of hydrologic information over each of the four seasons in California (CA) and the Great Plains (GP).

Fig. 6 shows that Q and R^- are the two dominant sources of hydrologic information to explain the uncertainty in reservoir releases. Furthermore, R^- is more important for weekly data than for monthly data while Q is more important during wet seasons than in dry seasons. Although Q^+ is on average the least important hydrologic variable, it explains roughly 21–33% and 18–33% of the uncertainty in decision making in California and the Great Plains, respectively.

Since past information is known with certainty while future information is forecasted and involves less certainty, variables that describe past hydrologic information (e.g., Q^-, S^-, or R^-) are expected to reveal stronger signals to release decisions than those describing future hydrologic information (e.g., Q^+). This is consistent with the results shown in Fig. 6 for the most part. Q^- and R^- are associated with greater average mutual information than Q^+ across all four seasons for both weekly and monthly datasets. S^- exhibits the same tendency of being more important than Q^+ on average except for the weekly data (reservoirs in California) in the spring season (flooding season). This indicates that future hydrologic information with larger uncertainty (e.g., Q^+) attains greater importance to decision making than some past hydrologic information (e.g., S^-) known with certainty during the flooding season in California. This does not hold true for longer temporal horizons (monthly) since the importance of Q^+ to release decisions diminishes with time due to lack of forecasting accuracy for long time horizons.

Comparing among the variables that describe past hydrologic information (e.g., Q^-, S^-, or R^-) reveals greater importance of R^-, on average, than Q^- or S^- to release decisions. One plausible explanation is that R^- is physically linked to the information contents of both Q^- and S^-. Knowing the amount of inflow received during the past period (Q^-), and the stage level of the reservoir at the end of last period (S^-) should end with a fair estimate of the amount of water released (R^-) during the past month (assuming that the uncertainty with losses from the reservoirs due to evaporation or seepage is negligible).

5.1. Statistical significance threshold

Based on the established $I_{\text{significance}}$ values for each reservoir and in each season, we determine which of the five sources of hydrologic information exhibits a statistical significance to reservoir releases for each of the selected reservoirs in California and the Great Plains. Fig. 7 shows that all five hydrologic variables are significant in more than 60% of the reservoirs. Although Q^+ is generally not the dominant variable, over 70% of the selected reservoirs in California (and 60% in the Great Plains) indicate a statistically significant relation between future inflows and release decisions. This strongly implies that reservoir operators rely on forecasted inflow at least partially in their decision process. Furthermore, it is intuitive that operators put more inference on deterministic knowledge such as past inflows, stage, and release than the uncertain future inflow. Also, Fig. 7 shows that a greater percent of reservoirs exhibit significant importance of hydrologic information in California than in the Great Plains. One may claim that more hydrologic information is utilized by operators in California than in the Great Plains. But, more conceivably, it is probably due to greater importance of hydrologic information to decision making with shorter horizons (e.g., weekly verses monthly). Hydrologic information during the past week or the next week is more likely to influence release decisions than their monthly counterparts.

5.2. Dominant source of hydrologic information

When reservoirs are classified by the single most important source of hydrologic information to release decisions, more pronounced trends emerge. Fig. 8 clearly shows that R^- and Q are the two most important hydrologic variables for the majority of the reservoirs in California and the Great Plains. In particular, the dominance of R^- is more evident for the weekly data over the monthly data, and Q gains a higher percentage during wet seasons. Furthermore, the importance of Q^+ to reservoir release decisions is shown to be significant (Figs. 6 and 7) although it is never the most significant variable in any of the reservoirs in California and is seldom in the Great Plains. Q^+ is the dominant variable in the Great Plains under two cases: for irrigation reservoirs when hydrologic uncertainty is relatively high and size ratio is relatively small; and for flood control reservoirs when hydrologic uncertainty is relatively low and size ratio is relatively large (see Table 1). The first case is plausible since higher uncertainty in inflow and smaller size reservoirs may require operators to pay more attention to long term predictions (e.g., monthly) to meet predicted farmers' demands. For the second case, all three reservoirs exhibit Q^+ as the strongest signal to release decisions (see Table 1) during the low

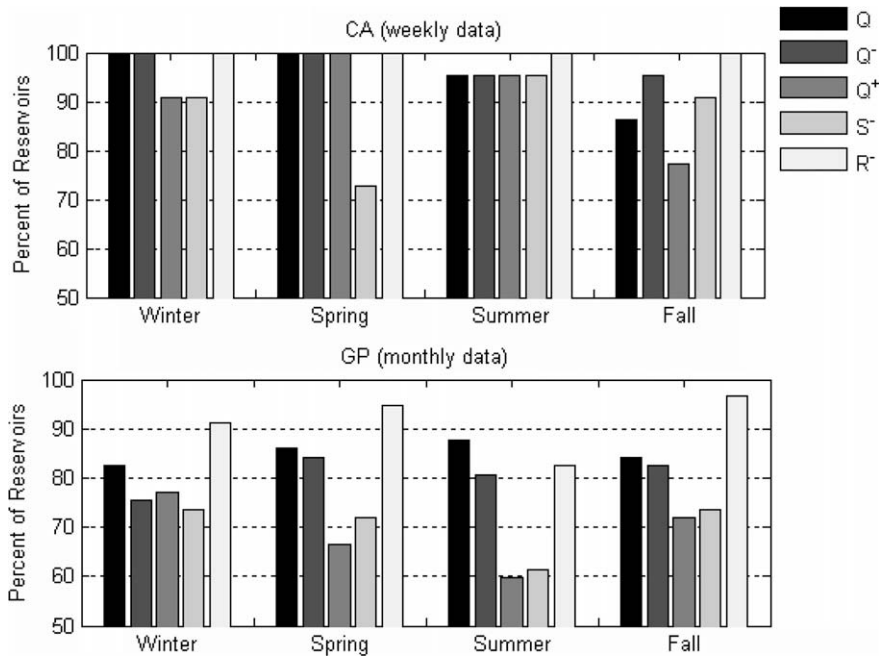

Fig. 7. Percent of reservoirs in California (CA) and the Great Plains (GP) that exhibit a statistically significant relationship to reservoir release decisions.

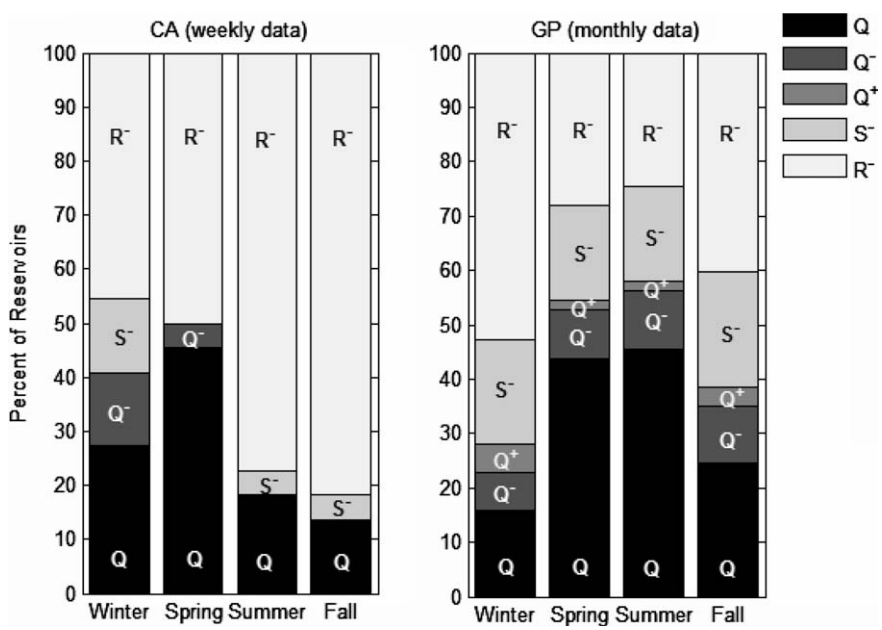

Fig. 8. Percent of reservoirs that have one of the five variables as the dominant variable of inference, based on 22 reservoirs in California (CA) and 57 reservoirs in the Great Plains (GP); high flow seasons are spring and summer in California and the Great Plains.

Table 1
List of reservoirs in the Great Plains region in which Q^+ exhibits the strongest signal to reservoir release decisions; seasons reflect those in which the dominance of Q^+ occurs

Reservoir name	State	Seasons	Reservoir use	Size ratio (%)	Hydrologic uncertainty	Statistically significant
Sherman Reservoir	NE	Spring / Fall	IR	43	2.60	Yes/Yes
Box Butte Reservoir	NE	Winter	IR	79	2.60	Yes
Nelson Reservoir	MT	Summer	IR	100	2.77	No
Bonny Reservoir	CO	Fall	FC	151	1.28	Yes
Foss Reservoir	OK	Winter	WS & FC	156	0.29	Yes
Webster Reservoir	KS	Winter	FC	680	0.67	Yes

FC: flood control; IR: irrigation; WS: water supply.

flow seasons (winter and fall) while the flooding seasons are mainly associated with the spring and summer seasons in the Great Plains region (Fig. 5). During the flooding seasons, short term forecasting or past hydrologic information may be more valuable to reservoir operators to avert from flooding but during the non-flooding seasons that role is no longer necessary. During the low flow seasons operators rather might be more interested in predicting how much water they anticipate during the flooding season to decide on how much water to store prior to the onset of the flooding season.

5.3. Reservoirs' main usage

When classifying reservoirs based on their primary purpose of use, a great difference between California and the Great Plains is observed. Table 2 summarizes the single most important source of hydrologic information for each four categories of primary reservoir uses (flood control, hydropower, irrigation, or water supply).

The operations of irrigation and water supply reservoirs in California are mainly tied to past weekly releases while monthly current inflow is the strongest in the Great Plains. Persistence and continuity are probably of great operational value to the operations of irrigation and water supply reservoirs at the weekly scale while at the monthly scale current inflow dictates decisions. It may also be that reservoirs with primary usage of irrigation or water supply are operated on a monthly scale with a constant monthly release function; a reason that may explain the great relevance of previous weekly release decisions. Interestingly, hydropower reservoirs show the opposite trend to irrigation and water supply reservoirs. At the shorter temporal scale, current inflow is a valuable piece of information to hydropower reservoir operators; but at the longer temporal scale, decisions are centered on maintaining a continuous hydropower production level. It seems that hydropower reservoirs rely on mixed information of immediate past and future flow conditions (e.g., the past and future few days) while monthly timescale is unlikely a time interval for the operation of those selected hydropower reservoirs. This should be consistent with the real-world situation.

Flood control reservoirs show mixed results across seasons. This is expected, because flood control operations resume during specific seasons only; thus, they should be analyzed in the context of flooding seasons only. In California, current inflow is the dominant source of hydrologic information in the flooding season (spring), which shows flooding control should be sensitive to the inflow conditions of immediate past and future few days. For instance, during the flooding season, if it rains heavily for the past couple of days or heavy storms are forecasted to occur over the coming couple of days, the reservoir operator would start releasing more water to cope with such high inflow conditions. For the Great Plains reservoirs, the monthly scale may be too coarse for flooding analysis as indicated by the mixed results.

Table 2
Summary of the most important source of hydrologic information on average when classifying reservoirs by their primary use

Reservoir use	California (weekly data)				Great Plains (monthly data)			
	Winter	Spring	Summer	Fall	Winter	Spring	Summer	Fall
FC	S^-	Q	R^-	R^-	R^-	S^-	Q^-	S^-
HP	Q	Q	Q	Q	R^-	R^-	Q	R^-
IR	R^-	R^-	R^-	R^-	R^-	Q	Q	Q
WS	R^-	R^-	R^-	R^-	Q	Q	Q	Q^-

FC: flood control; HP: hydropower; IR: irrigation; WS: water supply.

Table 3
Summary of the most important source of hydrologic information on average based on size ratios

Size ratio	California (weekly data)				Great Plains (monthly data)			
	Winter	Spring	Summer	Fall	Winter	Spring	Summer	Fall
Small	Q	Q	Q	R^-	Q	Q	Q	Q
Large	R^-	R^-	R^-	R^-	R^-	R^-	Q	R^-

5.4. Size ratio

Reservoir size ratio is a clear split on how reservoirs were operated. In general, large reservoirs (size ratio >100%) are identified with R^- as being the dominant source of hydrologic information (Table 3) while for small reservoirs (size ratio < 30%), Q emerges as the dominant source of information in both California and the Great Plains. Two exceptions are present in Table 3: in California, R^- is the dominant source of hydrologic information for both small and large reservoirs during the fall season, and in the Great Plains, Q is the dominant source of hydrologic information for both small and large reservoirs during the summer season. Recall, the fall season in California and the summer season in the Great Plains correspond to the lowest and highest flow seasons, respectively (Fig. 4). This can be explained by the fact that, with extremely low or high flow, reservoir operators are indifferent to whether the reservoir size ratio is small or large. Furthermore, for the fall season in California, R^- is much more important on average than the rest of the hydrologic variables (Fig. 6), statistically significant in all 22 reservoirs (Fig. 7) and the dominant source of hydrologic information in 82% of the reservoirs (Fig. 8). This implies that in the low flow season, the continuity of stable release might be the most important concern for reservoir operators. Similarly, for the summer season in the Great Plains, Q is the most important variable on average than the rest of the hydrologic variables (Fig. 6), statistically significant in 88% of the reservoirs (Fig. 7) and the dominant source of hydrologic information in 45% of the reservoirs (Fig. 8). This indicates that when the reservoir operation is viewed at a monthly time scale, the hydrologic information of a few weeks back and forth is most relevant to decision making.

In California, the importance of Q, Q^-, and Q^+ are strongly tied with reservoirs characterized by smaller reservoir ratios. Their importance to reservoir operations in the fall season decreases logarithmically with reservoir size ratio (Fig. 9). For the monthly data in the Great Plains, the importance of Q in the fall season decreases logarithmically with reservoir size ratio while Q^- and Q^+ show weaker signals in the fall season. Similar trends also appear in the spring and summer seasons; R^- and S^- exhibit no observable trend with reservoir size ratios. Additionally, decision uncertainty of the selected reservoirs in the Great Plains exhibits an inverse relationship with reservoir size ratio implying that monthly stochastic release decisions are more appropriate for relatively small reservoirs (Fig. 10).

5.5. Hydrologic uncertainty

Unlike reservoir size ratio, in general, reservoirs characterized with high hydrologic uncertainty (HU > 2bits) are identified with Q being the dominant source of hydrologic information (Table 4) while for reservoirs with low hydrologic uncertainty (HU < 1 bits), R^- becomes the dominant source of information in both California and the Great Plains. As indicated in Table 4, in the winter and summer seasons in California and the fall season in the Great Plains, in which both low and high flow are possible, Q is the dominant hydrologic information with high hydrologic uncertainty, while R^- with low hydrologic uncertainty. However, for the seasons in which regular low or high flow occurs, hydrologic uncertainty does not appear as a sensitive factor. For example, in the

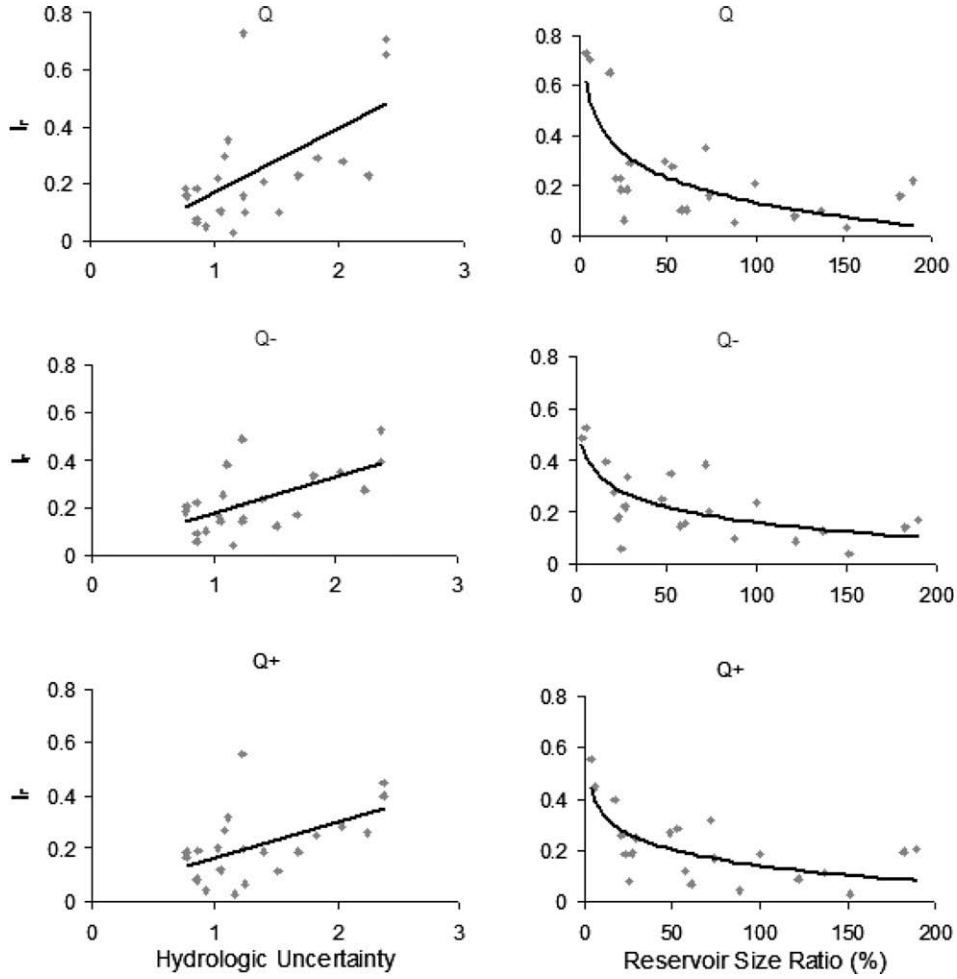

Fig. 9. In California, the importance of Q, Q^-, and Q^+ increases linearly with hydrologic uncertainty and decreases logarithmically with reservoir size ratios approximately (based on the data of the fall season only).

dry seasons, i.e., fall in California and winter in the Great Plains, R^- is the dominant source of hydrologic information for reservoirs with both low and high hydrologic uncertainty. In wet seasons, i.e., spring in California and spring and summer in the Great Plains, Q is the dominant source of hydrologic information for reservoirs with both low and high hydrologic uncertainty. Thus, the magnitude of inflow (represented by multiple year average) and the level of uncertainty affect reservoir release jointly.

Table 4
Summary of the most important source of hydrologic information on average based on hydrologic uncertainty

Hydrologic uncertainty	California (weekly data)				Great Plains (monthly data)			
	Winter	Spring	Summer	Fall	Winter	Spring	Summer	Fall
Low	R^-	Q	R^-	R^-	R^-	Q	Q	R^-
High	Q	Q	Q	R^-	R^-	Q	Q	Q

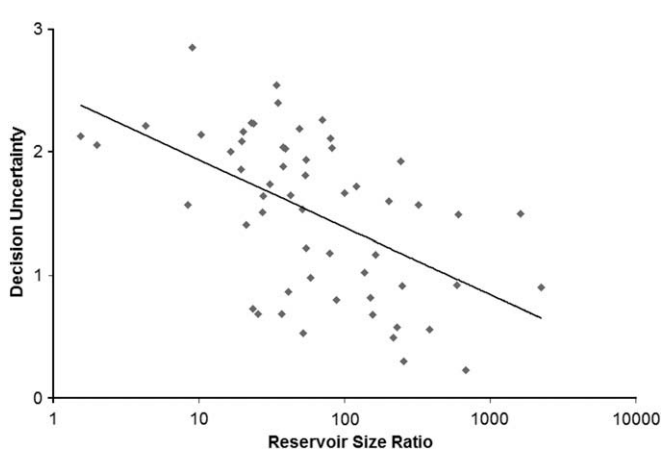

Fig. 10. Reservoir size ratio vs. decision uncertainty for the Great Plains (GP) exhibits an inverse relation between the two variables.

Reservoirs characterized with high hydrologic uncertainty at the weekly scale tend to rely more pronouncedly on the three inflow-related sources: Q, Q^- and Q^+ (Fig. 9). Their importance to reservoir operations in the fall season increases approximately linearly with hydrologic uncertainty. At the monthly scale, in the Great Plains, that tendency diminishes except for Q^+ which shows a positive relation to hydrologic uncertainty over all spring, summer and fall seasons. R^- and S^- exhibit slightly greater importance with lower hydrologic uncertainty in the Great Plains while only S^- in California.

Another interesting result is the linear association between hydrologic uncertainty (randomness of reservoir inflows) and the decision uncertainty (randomness of reservoir releases) presented in Fig. 11. The relationship is somewhat intuitive. The more

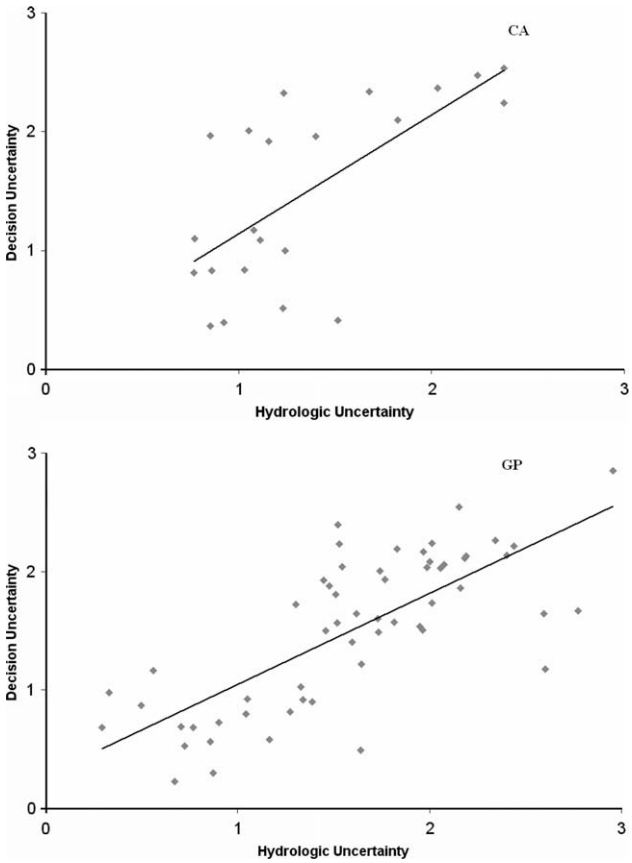

Fig. 11. Hydrologic uncertainty vs. decision uncertainty for both California (CA) and the Great Plains (GP) exhibits a proportionally linear relation between the two variables.

uncertain the inflow, the more uncertain the operator decision will be. Slope of the two lines is similar, which may imply that reservoir operators in California and the Great Plains behave similarly in response to hydrologic uncertainty.

5.6. Average annual precipitation

Splitting the reservoirs based on those located in wet regions and those in dry regions has revealed that more hydrologic information is associated with the dryer regions in California and with the wetter regions in the Great Plains. Fig. 12 shows the difference in the importance of the individual sources of hydrologic information in both wet and dry regions (e.g., $I_r^{\text{wet}} - I_r^{\text{dry}}$). So a positive value indicates that hydrologic information is of greater importance in wet over dry regions. Hence, hydrologic information is of greater importance to reservoir release decisions in dry (over wet) regions in California and in wet (over dry) regions in the Great Plains. In California, reservoirs in wet regions are mainly located in northwestern California while reservoirs in dry regions are mainly in the central valley and southern region of California. The existence of a north–south water demand phenomenon places reservoirs in wet regions under intense demand for water supply to dry regions in California. Therefore, in California, reservoir operators in wet regions may rely more in their decision process on other factors such as projected water demands in the south over the five hydrologic variables considered in this study; while in dry regions, local precipitation is very low and operators are more sensitive to the hydrologic parameters of reservoir inflow and storage, as examined in this study. In the Great Plains, that north–south demand phenomenon does not exist and water shortage is less extreme than that in dry regions in California.

Another plausible explanation is that the reservoirs in wet regions in California and dry regions in the Great Plains rely more on other sources of hydrologic information (e.g., snowpack and snow melting, antecedent soil moisture conditions, farmers' demands, etc.) than by those considered in this study (Q, Q^-, Q^+, S^-, or R^-). In California, for example, reservoirs classified as in wet regions receive their inflow mainly through snow melting; thus, decision making may be more influenced by the amount of accumulated snowpack from the past winter season than the hydrologic variables considered in this study.

5.7. Individual reservoirs

In Fig. 13 we present six individual reservoir examples that illustrate the three common trends of the most important source of hydrologic information in each of the four seasons to decision making in both regions. For the first group, Q is the dominant source across all four seasons such as shown by Whiskeytown Dam, CA and Anchor Reservoir, WY. For the second group, R^- is the main source of hydrologic information as depicted by Trinity Dam, CA, and Pathfinder Reservoir, WY. The third group is when Q is dominant during wet seasons and R^- during dry seasons (less evident) as illustrated by Hidden Dam, CA, and Shadehill Reservoir, SD. When investigating the average size ratio and average hydrologic uncertainty associated with each group, it is found that the first group is mainly associated with small reservoirs and those with higher hydrologic uncertainty. The other two groups are linked mainly to large and hydrologically uncertain reservoirs (Fig. 14).

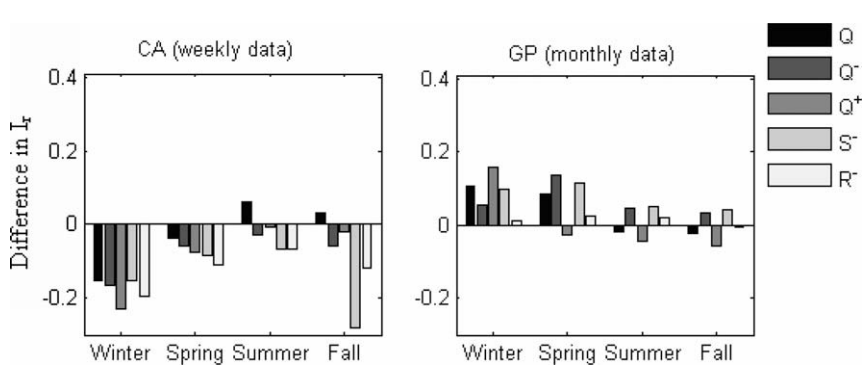

Fig. 12. The difference (wet minus dry) in average relative mutual information when classifying reservoirs in California (CA) and the Great Plains (GP) based on their average annual precipitations.

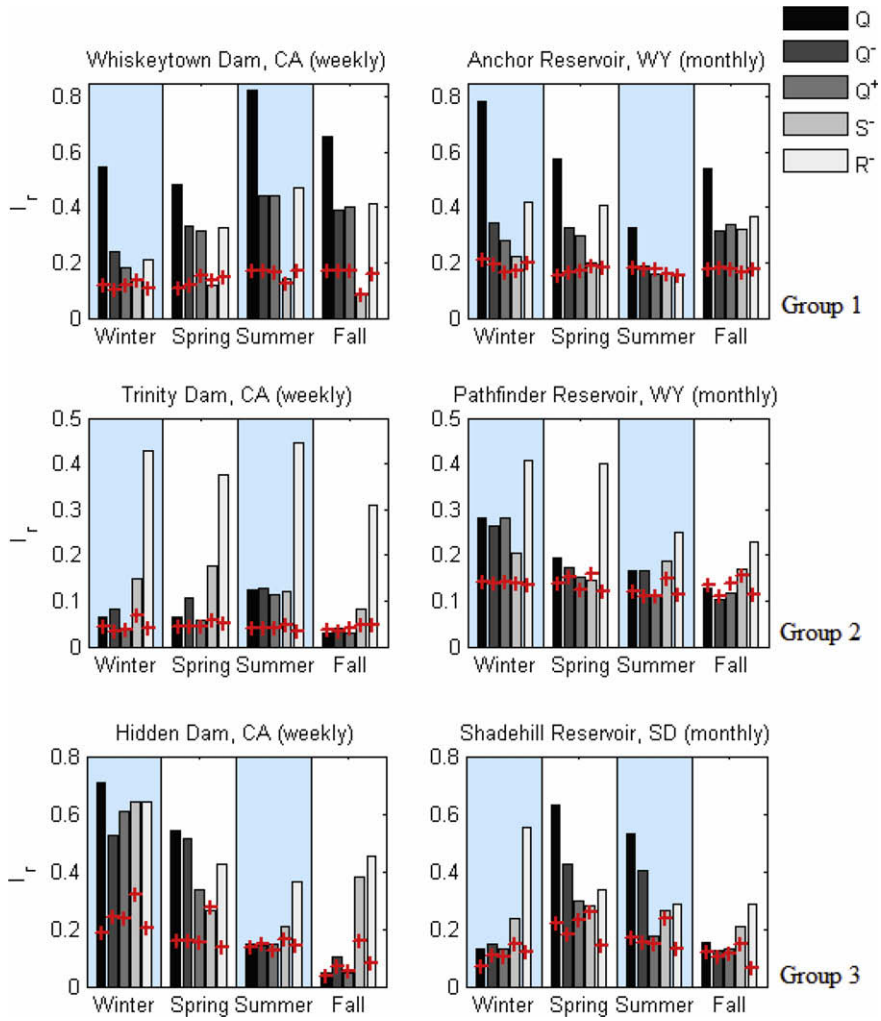

Fig. 13. Selected individual reservoirs (3 in California and 3 in the Great Plains); the '+' marks indicate the statistical significance threshold values ($I_{significance}$) with a 99.5% significance level; Group 1: Q is the dominant source of hydrologic information across all four seasons; Group 2: R^- is the main source across all four seasons; Group 3: Q is dominant during wet seasons and R^- during dry seasons.

6. Discussion

6.1. When is Q more important than Q^- or Q^+?

Although it is shown that in general Q is a more important source of hydrologic information to reservoir operators than Q^- or Q^+, there are some exceptions. Thus, we further discuss the question of when Q is less (or more) important than Q^- and Q^+. At first glance, it can be seen that reservoirs with $Q > Q^-$ and $Q > Q^+$ are associated with smaller reservoirs, and $Q < Q^-$ and $Q < Q^+$ are associated with larger reservoirs (Fig. 15). This may sound contradictory for California in particular and more specifically to the results shown in Fig. 9, which shows that all of Q, Q^-, and Q^+ decrease logarithmically with reservoir size ratio. However, it is important to point out that Q^- and Q^+ are more important than Q mainly when the importance of Q is relatively low (Fig. 16). In other words, Q is generally more important than Q^- or Q^+ to reservoir operators, and Q^- or Q^+ gain greater importance to release decisions solely when the relative importance of Q is low. This implies that when Q is less important, more consideration might be given to Q^- and Q^+ for larger reservoirs.

During the flooding seasons, however, Q^+ is more important than Q when hydrologic uncertainty is high for both the weekly and monthly datasets (Fig. 17). Therefore, only during the flooding seasons and when hydrologic uncertainty is high, forecasted inflow (Q^+) reveals greater importance to decision making than current inflow (Q).

6.2. The impact of weekly and monthly time intervals

Reservoir operators exhibited differing decisions' behaviors in response to hydrologic information and hydrologic uncertainty between California and the Great Plains (e.g., Table 2). More realistically, such observation may have to do with the variation in time horizon between the weekly (for California) and monthly (for the Great Plains) datasets. To investigate this further we compare the two temporal scales (weekly and monthly) for Whiskeytown Dam in California. We run the weekly data 100 times with the same sample size of the monthly data but drawn randomly to isolate any data size effect on the results. Table 5 indicates that only Q is of greater importance at the longer time horizon while the importance of the rest of the variables decrease with longer time horizons. This shows that immediate past and future hydrologic information is more important for reservoir release decisions. Acquiring hydrologic information about the conditions of the reservoir system over the past week is probably more important to release decisions than over the past month. Similarly, Q+ over the coming week is definitely more reliable than over a month period.

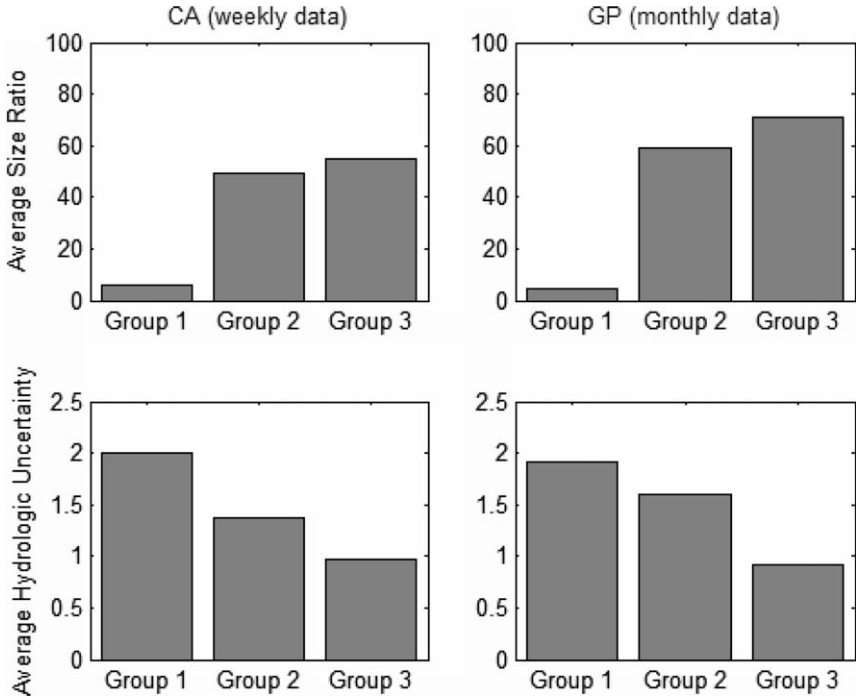

Fig. 14. Average hydrologic uncertainty and reservoir size ratios for each of the three identified groups in California (CA) and the Great Plains (GP); Group 1: Q is the dominant source of hydrologic information across all four seasons; Group 2: R^- is the main source across all four seasons; Group 3: Q is dominant during wet seasons and R^- during dry seasons.

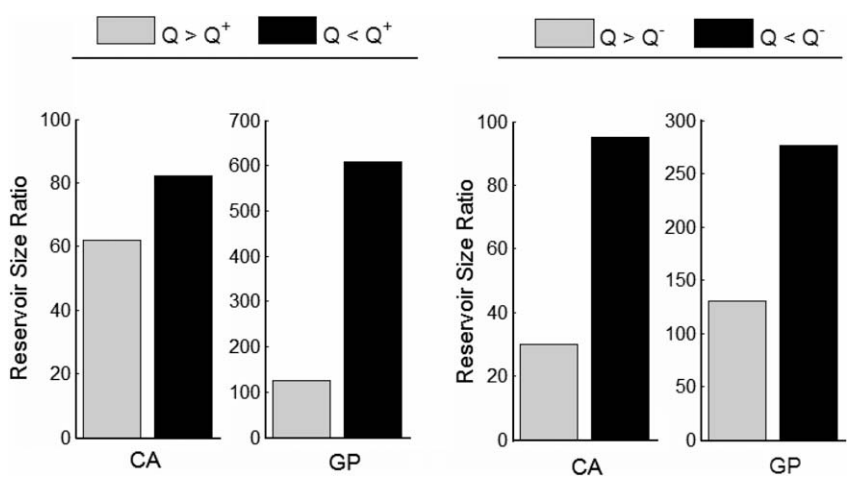

Fig. 15. Comparing average reservoir size ratio when current inflow is more important than future inflow ($Q > Q^+$) or past inflow ($Q > Q^-$), and when current inflow is less important than future inflow ($Q > Q^+$) or past inflow ($Q > Q^-$); Q^- and Q^+ are more important than Q when reservoirs are relatively large; data shown for summer season only.

In the case of Q, for shorter intervals (e.g., weekly), the content of information is probably limited and reservoir operators may inference their decisions on further future and past inflows (e.g., next and last weeks' inflows). However, for longer intervals (e.g., monthly), the content of information during the past or future months is probably less valuable to release decisions than that during the current month and more inference is then based on the current inflow period.

6.3. Implications for building more realistic reservoir optimization models

In this paper, we foresee more realistic reservoir model formulations by learning from past releases by investigating historical data from 79 reservoirs in California and the Great Plains regions in the US. The ultimate target of this study is to advance more efficient optimization techniques that enable practitioners to formulate the reservoir optimization problem without oversimplifying necessary details of the system dynamics. Specific discussions are provided as follows.

Classic reservoir optimization models seek the establishment of an 'optimal' reservoir policy based on a predefined objective function (e.g., maximizing benefits, etc.) and a set of constraints. The objective function is simply a mathematical representation that should reflect and dictate the goal of the reservoir operator. Yet, reservoir operators often behave differently in the face of various hydrologic conditions. Lack of sufficient hydrologic information in the understanding of the sensitivity of reservoir operators to cer-

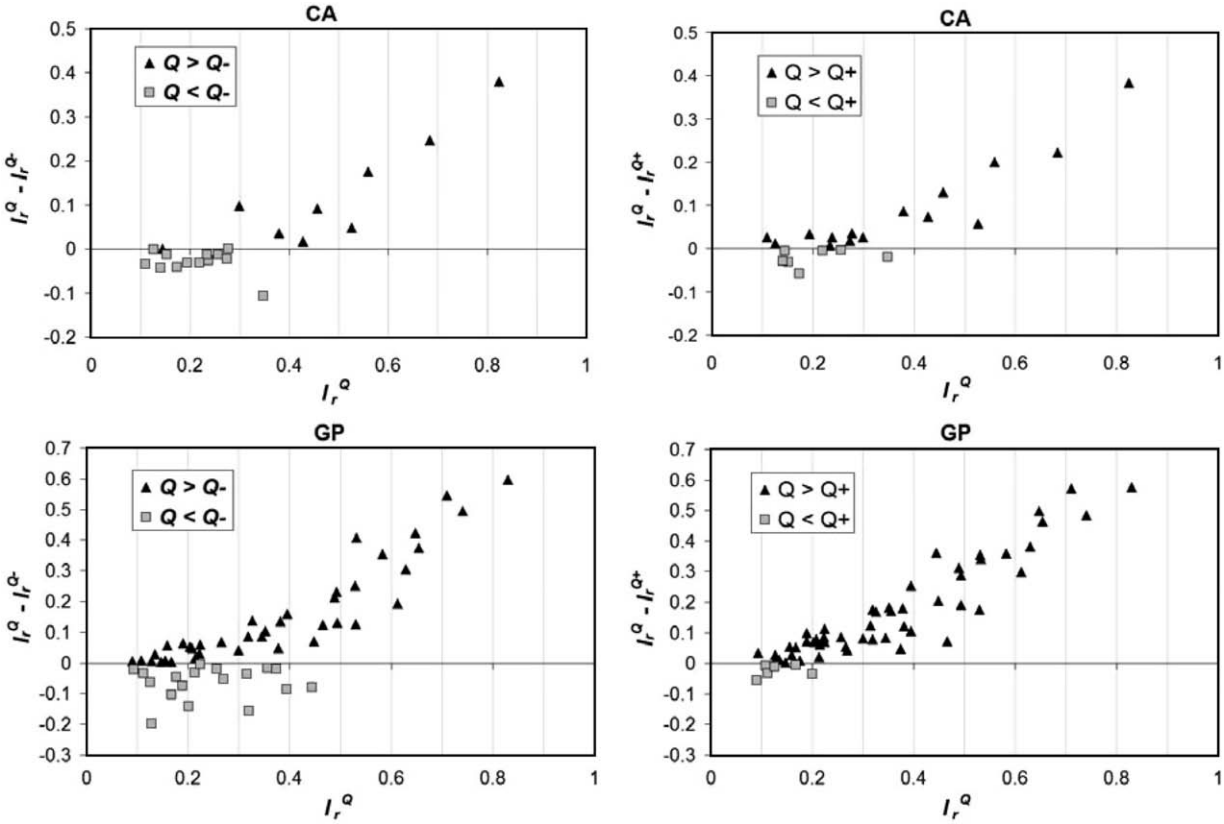

Fig. 16. Difference in relative mutual information between Q and Q^- plotted against the mutual information of Q; positive and negative points correspond to when $Q > Q^-$ (or $Q > Q^+$) and when $Q < Q^-$ (or $Q < Q^+$), respectively; Q^- and Q^+ are more important than Q when I_r^Q is relatively small; only summer season data is shown; top: reservoirs in California (CA); bottom: reservoirs in the Great Plains (GP).

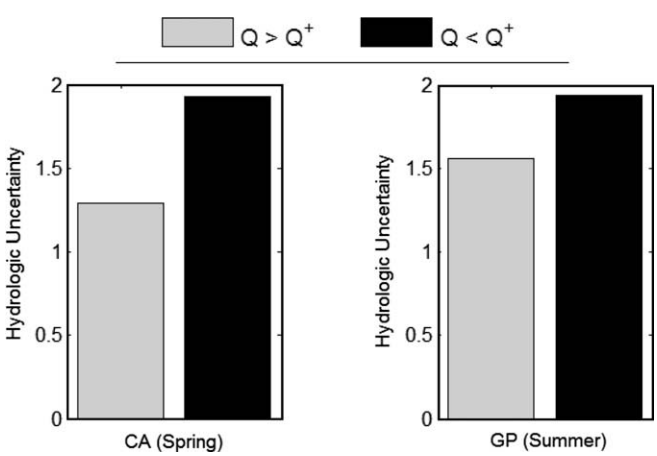

Fig. 17. Comparing average hydrologic uncertainty when current inflow is more important than future inflow ($Q > Q^+$), and when current inflow is less important than future inflow ($Q > Q^+$); Q^+ is more important than Q when reservoirs are characterized with low hydrologic uncertainty during the flooding seasons in California (CA) and the Great Plains (GP).

tain hydrologic conditions in the model formulation will lead to unrealistic representation of the driving forces for reservoir release decisions.

Extracted from historical reservoir release data, it is shown that reservoir operators are sensitive to reservoir size and hydrologic uncertainty, especially the inflow-related hydrologic information. The value of past, current, and future inflow information to release decisions decreases logarithmically with reservoir size and increases linearly with hydrologic uncertainty. In other words, the value of inflow conditions to reservoir operators is greater when the reservoir size ratio is small and the hydrologic uncertainty of inflow is high. For small reservoirs, the amount of annual inflow into the reservoir greatly exceeds the maximum capacity of the reservoir; thus, the reservoir is associated with higher risk of flooding. Also, when reservoirs are characterized by high hydrologic uncertainty, reservoir operators are dealing with more unpredictable inflows. Therefore, reservoir release decisions are related to reservoir size ratio and hydrologic uncertainty. This finding reiterates the claim by Labadie [13] that establishing reservoir optimization models that better off incorporate hydrologic uncertainty and risk is a necessity to close the gap between models and the real world.

Table 5
Comparing the relative importance of each of the five sources of hydrologic information using both weekly and monthly datasets for Whiskeytown Dam in California

Temporal scale of data	Hydrologic variables				
	Q	Q^-	Q^+	S^-	R^-
Weekly data	0.55	0.32	0.31	0.16	0.34
Monthly data	0.63	0.22	0.22	0.12	0.19

Moreover, the seasonal data classification reveals that reservoirs are operated differently over different seasons and the value of hydrologic information changes from one to another. The dominant source of hydrologic information to decision making changes across seasons. For instance, current inflow is the most important hydrologic variable to current release for reservoirs with high hydrologic uncertainty but forecasted inflow is of greater

importance than current inflow during flooding seasons. Additionally, current inflow is more important in high flow seasons while past release is more important during low flow seasons. Therefore, reservoir operators tend to follow different operation policies during the four seasons; thus, a more realistic reservoir optimization model should capture this by establishing a policy or set of policies per season.

Traditionally, dynamic programming (DP) and stochastic dynamic programming (SDP) reservoir models are limited to an individual hydrologic variable (e.g. Q, Q^-, or Q^+), but the results from this study clearly show that such models may need to incorporate more hydrologic information in their formulations beyond the inflow condition during the previous time interval. Both past and forecasted inflow conditions, for example, may be essential to more adequately reflect the release decision process. Lack of incorporating all the hydrologic information influencing the decision behaviors of the reservoir operator may lead to formulating inadequate reservoir optimization models that are far from the real world. Furthermore, the authors find that no single hydrologic variable is dominant across all reservoirs in either of the two regions. Thus, researchers need to be cautious to the selected hydrologic variable to use in a DP or SDP formulation. The hypothesis here is that if such optimization models utilize more hydrologic information, they become closer to the real-world operations. This needs further investigation following this study.

The results also indicate that forecasted inflow is of great importance to reservoir release decisions under certain circumstances such as high hydrologic uncertainty and low reservoir size ratio. Therefore, researchers need to incorporate hydrologic forecasting models to attain a more realistic representation of the decision process. As illustrated by Figs. 6–9, the importance of a particular source of hydrologic information (e.g., Q^+) to release decisions may change dramatically across seasons or even more pronouncedly across different reservoirs with various characteristics (e.g., use, size ratio, hydrologic uncertainty, etc.). Thus, a realistic advancement is to establish the dominant sources of hydrologic information to reservoir operators prior to the modeling step to ensure that the most valuable hydrologic information is incorporated in the reservoir optimization formulation.

In a traditional reservoir model, decision and hydrologic information horizons are specified as prior knowledge to the model formulation step. For example, the modeler needs to establish whether to take decisions at the weekly, monthly, or even longer scales. Also, a hydrologic variable is typically used at the same temporal scale of the decision horizon. However, data may reveal that release decisions are taken at different temporal scales and hydrologic information used by operators is over longer or shorter horizons than the decision's horizon. Therefore, investigating explicitly and systematically the extents of decision horizon and of hydrologic information horizon from a data perspective may be essential; this is still unaddressed in this study and will require further research.

7. Conclusions

The approach provided in this paper extracts nonlinear ties among hydrologic variables. It enables modelers to extract important knowledge from historical release records about reservoir operators' behaviors and to investigate the influence of factors such as reservoir size ratio, hydrologic uncertainty, reservoir usage, and climate in the drainage area, on the value of hydrologic information to reservoir release decisions. These methods may be used as a discovery tool to identify which hydrologic variable or set of variables to use in a reservoir optimization model. The data-mining approach presented in this study is recommended as a complement to traditional modeling.

Our results indicate several interesting relations between historical releases and hydrologic information. Although all five sources of hydrologic information are significantly tied to the decision process of the majority of the 79 reservoirs, operators in California and the Great Plains relied mainly on inflow in the current time period (which is a mixture of immediate past and future information, Fig. 4) and previous releases to guide their operation decisions. The value of forecasted inflow in California increases linearly with hydrologic uncertainty and decreases logarithmically with reservoir size ratio. Yet, no single hydrologic variable is consistently dominant across all reservoirs and the value of hydrologic information varies seasonally. The significance of hydrologic information is sensitive to hydrologic uncertainty and reservoir size ratios, as well as to reservoir primary uses.

Two findings of this study confirm the findings of Tejada-Guibert et al. [33]. First, more complete hydrologic information may lead to better performance. Second, current inflow is shown to be more important to reservoir operators' decisions than other sources of hydrologic information such as past inflows and forecasted inflows. In this study, we show that incorporating more hydrologic information more appropriately explain reservoir release conditions; and current inflow is generally more reflective of release decisions than past or forecasted inflows. The difference between this work and that of Tejada-Guibert et al. [33] is that they evaluated information sources based on a set of predefined objective functions while we arrived at the same conclusions using real-world historical data. The results also confirm the findings of Karamouz and Houck [9]. They concluded that deterministic dynamic programming is more effective for medium to large reservoirs while stochastic dynamic programming is more effective for small reservoirs. In other words, it might be more necessary to incorporate stochastic information for relatively small reservoirs. In this study, we show that small reservoirs are characterized with high decision uncertainty (Fig. 10).

In the real world, operators usually follow prescribed operation rules, which to a large extent depend upon different operational objectives and practices such as flood control, water supply and hydropower generations; the decisions may also be driven by prompt or projected future water demands. In this study, we do not simulate reservoir operators' decisions on releases but rather explore the correlation between releases and a set of hydrologic variables, using a data-mining approach. Thus, our current work does not explicitly consider the impact of operation rules and reservoir functions. However, the impact is implicitly accounted for in the data-mining analysis; as shown in Table 2, the significance of hydrologic information is sensitive to reservoir primary uses. A valuable future research would be to incorporate the operation rules and other driving forces explicitly into the data-mining procedure as a "guidance" for mining the historical data.

Another issue to deal with in the entropic methodology presented herein is to differentiate between hydrologic inflow variability and uncertainty. Both Shannon entropy and mutual information used in this paper essentially represent the uncertainty involved in the data sets, while the latter reflects to what degree the variance (or information) with one data sets affects the variance (or information) of another data set. However, inflow variability, for example, seasonality can be "misunderstood" as "uncertainty" if the variability, which is usually predictable, is not identified separately from the uncertainty. Our study simply divides the data sets by season (e.g., winter, spring, summer, fall) to filter the impact of inflow variability. Actually, the operation rules and release demands are usually driven by inflow variability. As mentioned above, for future work the operation rules can be used to separate variability and uncertainty in the data-mining analysis.

A third issue related to the analysis presented in this paper is the role of forecast inflow in operators' decision making. Any forecast involves uncertainties and those can be provided as probabilistic forms. Our study does not consider the forecast uncertainty explicitly. For future research, obtaining knowledge on forecast uncertainty can help data mining to extract out more useful information about the behavior of operators in face of forecasted hydrologic uncertainty.

Additionally, we establish dependence of reservoir release decisions on each of the five sources of hydrologic information individually. However, dependence among the five sources is likely to exist, in the form of overlapping information contribution. Thus, we can state which source of hydrologic information is most important to reservoir release decisions, but we cannot determine which combination of multiple variables is most important using the present methods. To accomplish such a task, one would have to account for any dependence among the five hydrologic variables to avoid double-counting of predictive information. Computation of multi-dimensional conditional mutual information is required to account for the overlapping contribution of five different variables. Such computations are theoretically possible, but would require a number of data far larger than what is available in historical weekly or monthly reservoir time series datasets.

The data-mining results from this study might be useful for enhancing traditional optimization models for reservoir operations. More knowledge about how reservoirs were operated in the past may allow the research community to close the gap between the theoretical efforts and real-world implementations. In the future, such discoveries may help establish more optimal reservoir operation policies. Further research needs to be directed to establish the temporal extent of valuable hydrologic information solely from a data perspective. Specific opportunities may include investigating finer temporal scale release data to account for any dependence among hydrologic variables, or expanding the scope of the study by incorporating more comprehensive sources of hydrologic information and more reservoirs from different hydroclimatic regions.

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